Road Network Description File Annotation

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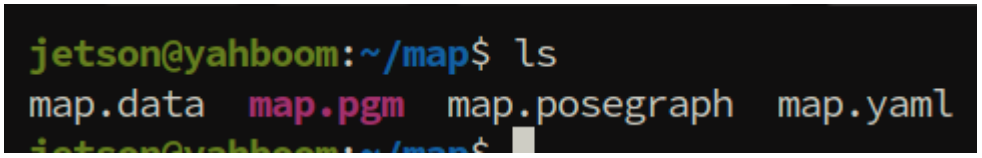
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1. Course Content

- Use the `route_tool` utility to annotate the road network and generate a `.geojson` file that describes the road network

2. Preparation

- Refer to the LiDAR chapter 【slam-toolbox Algorithm Mapping】 to build a **grid map** and a **pose graph format pose map**. After mapping is complete, the `$HOME/map` folder should contain the following 4 files:



2.1 Starting the Road Network Annotation Program

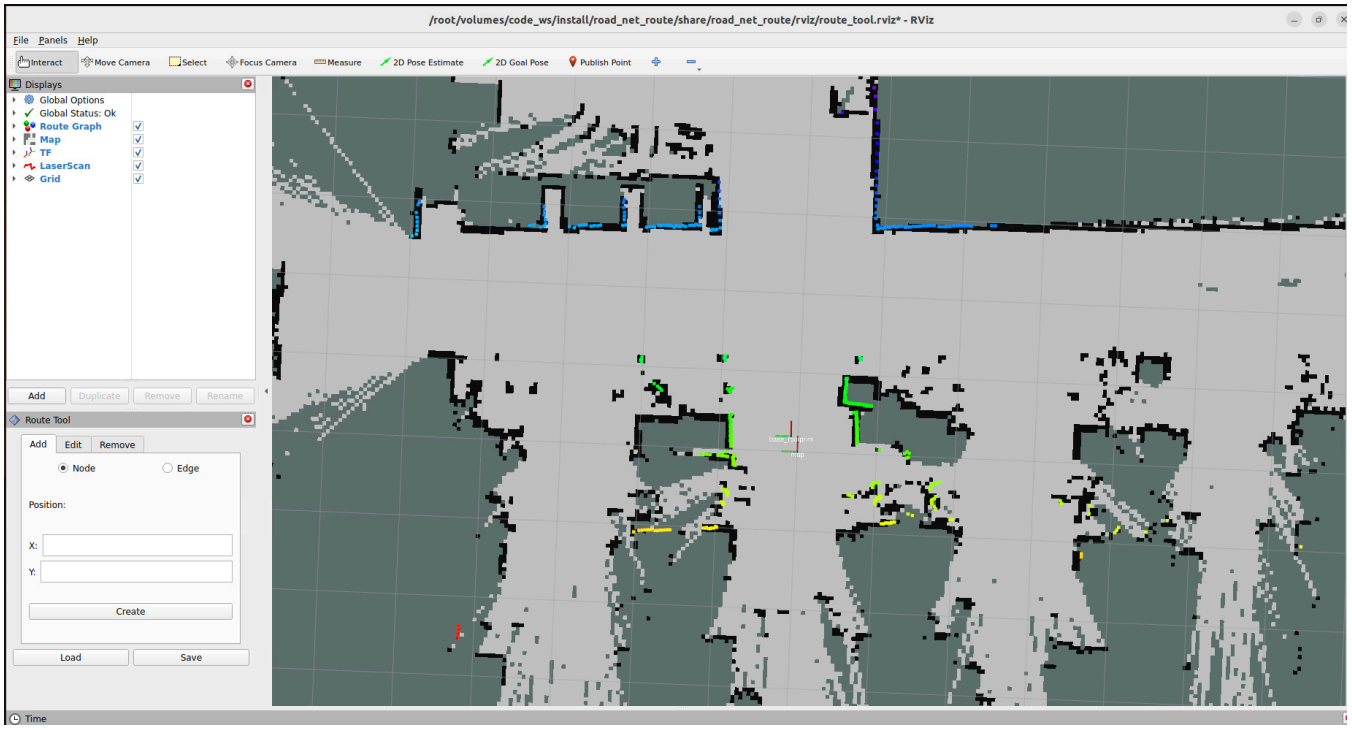
- Method 1: Start via **RobotManager**



- Method 2: Start via the CLI command line

```
ros2 launch yahboomcar_bringup nav2.launch.py rviz_config:=roadnet_edit
localization_mode:=slam_toolbox
```

- RViz will display the map and two coordinate frames, where `map` is the map coordinate frame, and `base_footprint` is the coordinate frame of the projection point on the ground at the center of the robot's chassis. Through `base_footprint`, we can know the robot's exact position in the map in real time.



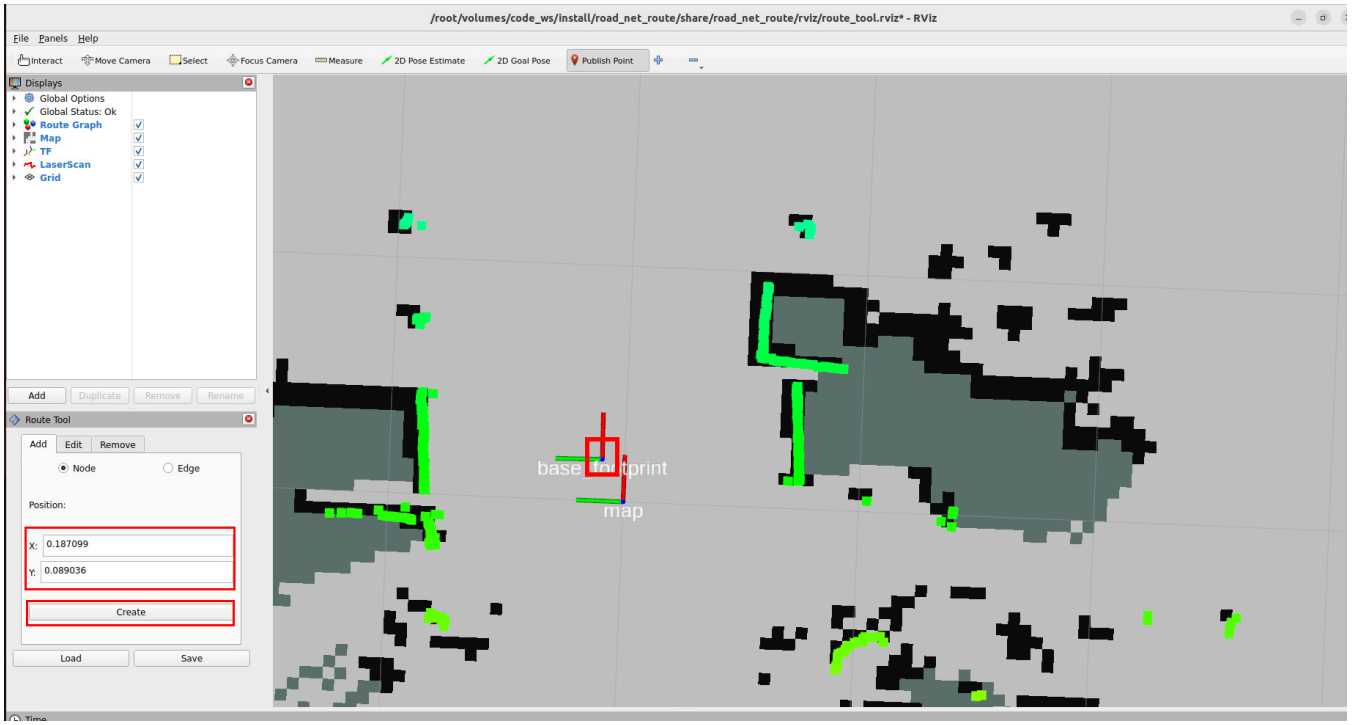
3. Road Network Annotation Case Demonstration

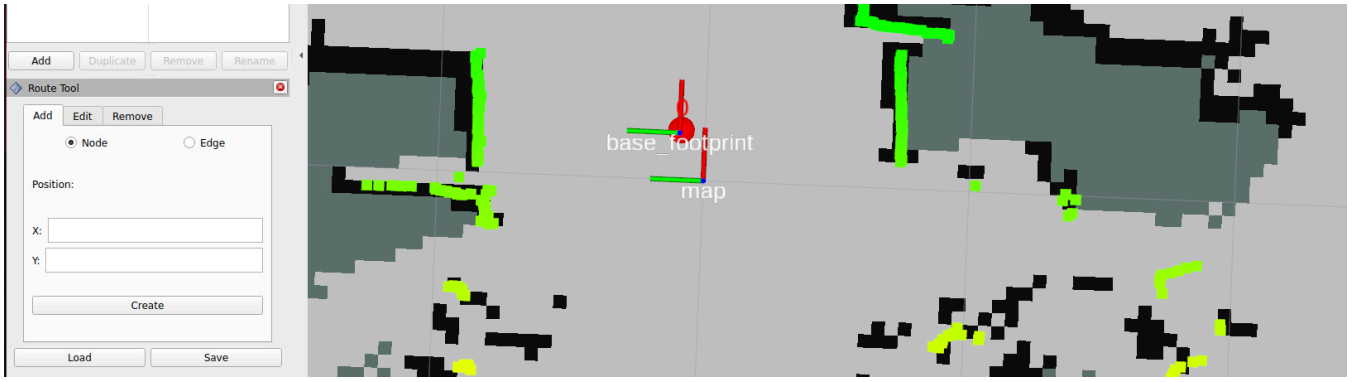
3.1 Adding Waypoints

- Click the `Publish Point` tool in the RViz toolbar to enter the waypoint selection state



- Then left-click the mouse at the small blue dot at the center of the `base_footprint` coordinate frame. The `position` field under the `Route Tool` tab at the lower left will automatically fill in the coordinates of the current point, and then click Create to create a waypoint.

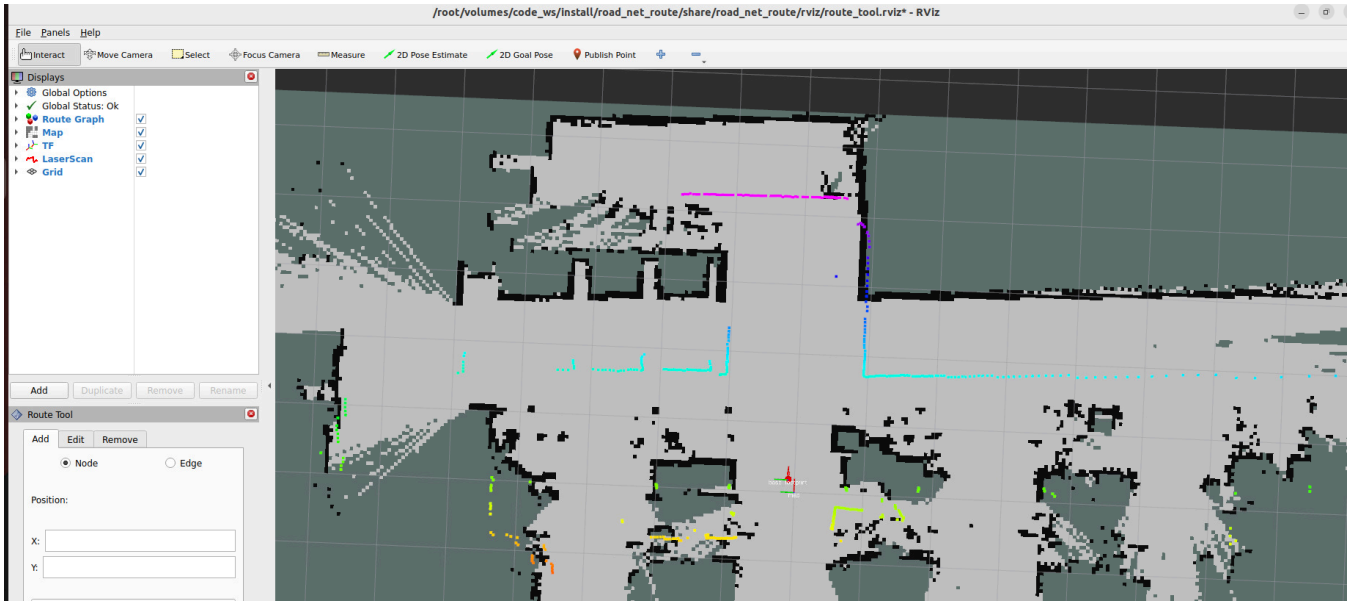




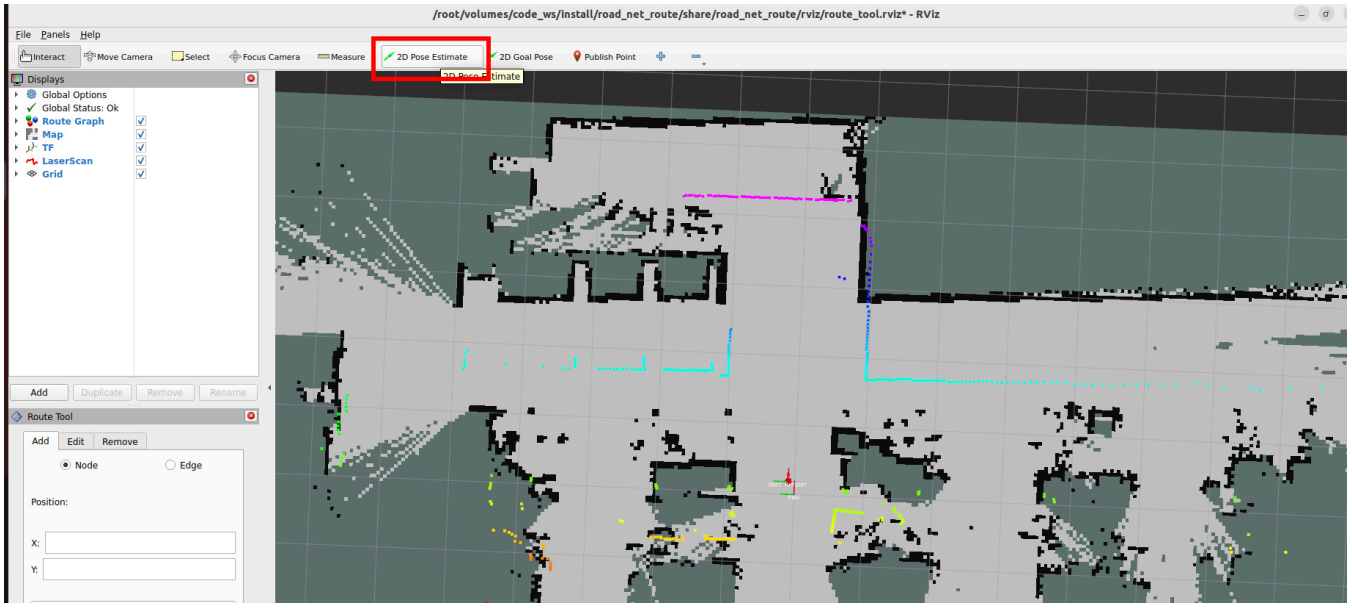
- Next, we move the robot to the next position where a waypoint needs to be marked. What is demonstrated here is directly carrying the robot to the next position where a waypoint needs to be marked.

> **Tip**
>
> Here the robot can be moved by keyboard control, joystick control, or `2D Goal Pose` navigation in RViz, or it can even be carried directly to the next position to be marked, because the pose map allows the robot's global position to be quickly and accurately localized.

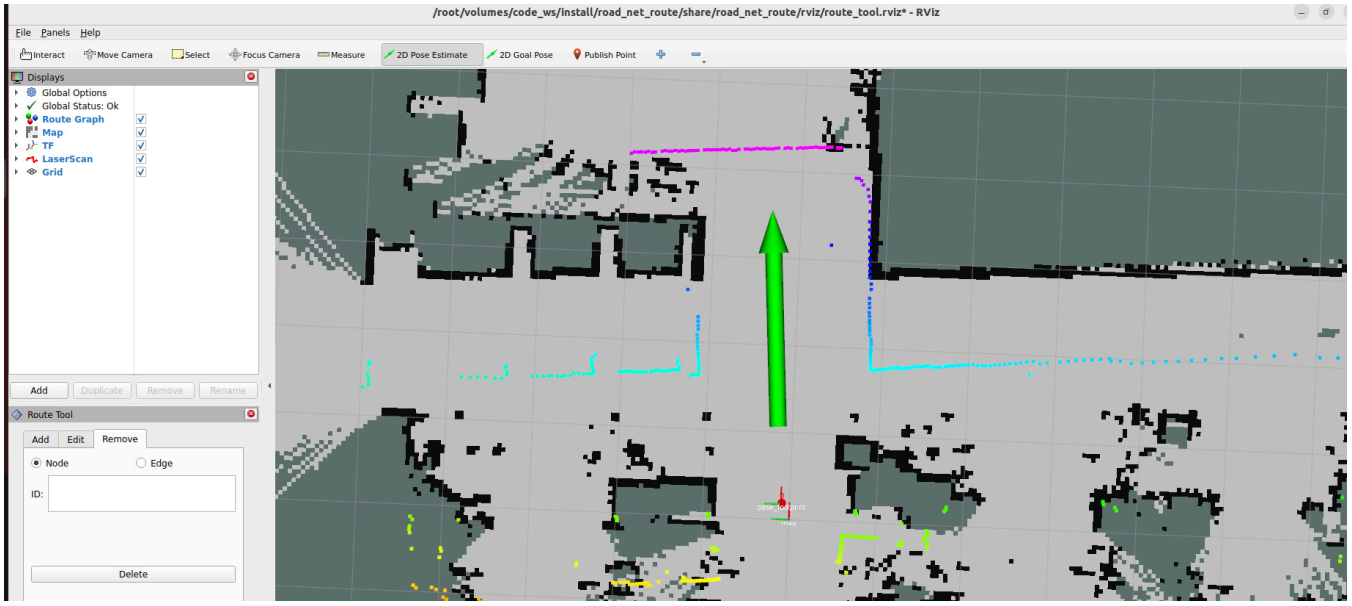
- As you can see, because the robot was moved directly, the global localization is lost, and the LiDAR contour no longer matches the map's environmental features. Here we can use the pose map to perform fast relocalization.



- Click the `2D Pose Estimate` tool in the RViz toolbar

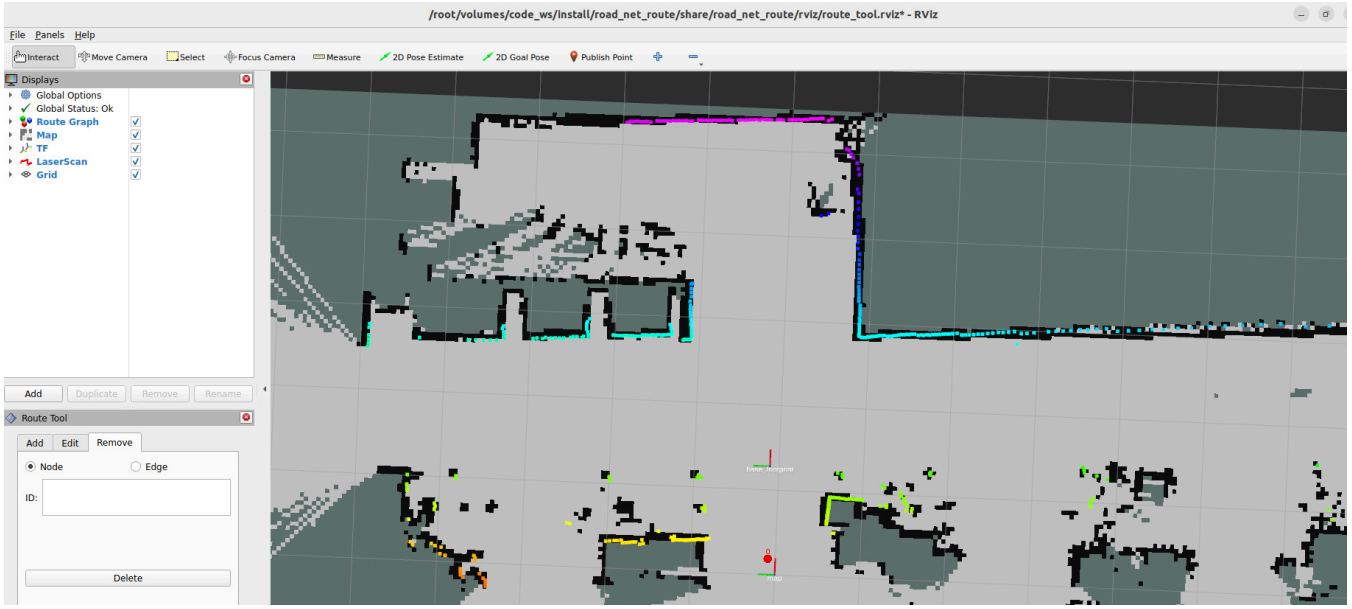


- Then give an estimated position near the robot's approximate position

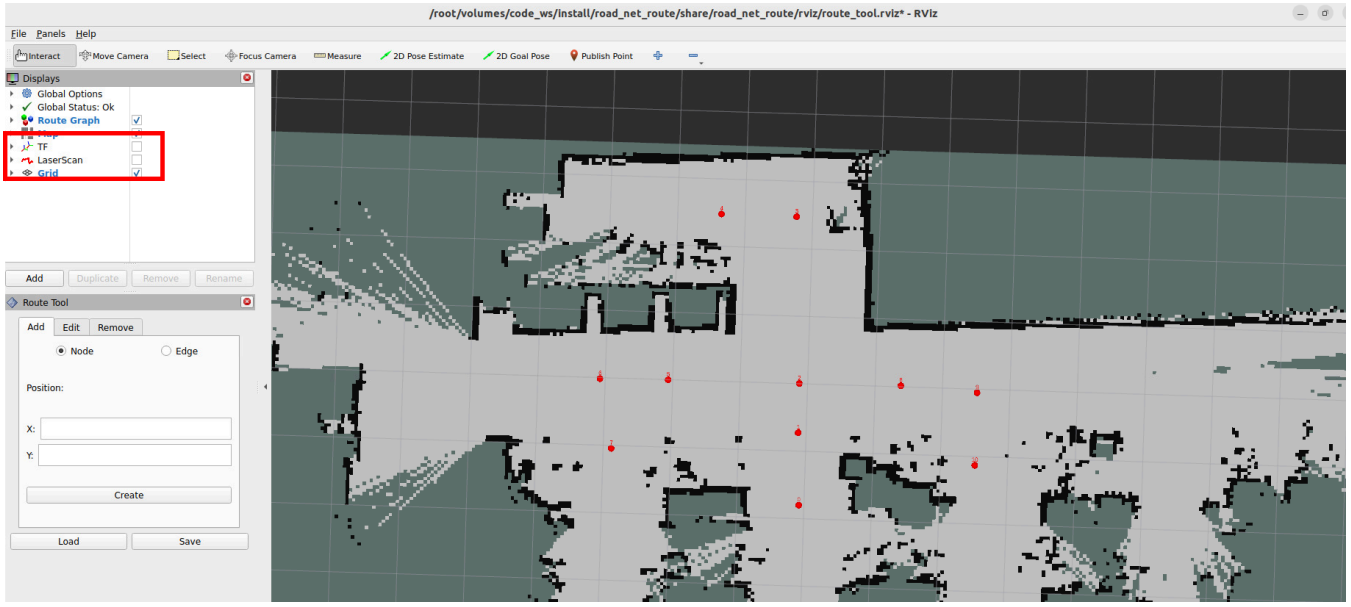


- After that, the robot's exact localization in the global map will be restored immediately

> **Tip**
>
> - The principle of this fast relocalization is to use slam_toolbox pure localization and the pose map to perform fast scan-map matching.
> - **Note:** During the waypoint marking process, if you observe that **the LiDAR contour deviates from the map's environmental features** (cumulative odometry error, wheel slip, manual movement, and various other factors can all cause local localization errors), you can use this method to quickly restore the robot's exact localization.



- Repeat the above steps until all the waypoints that need to be marked are completed (mark waypoints according to the user's own actual needs)
- After marking is complete, you can first turn off TF and LaserScan on the left to check the waypoints



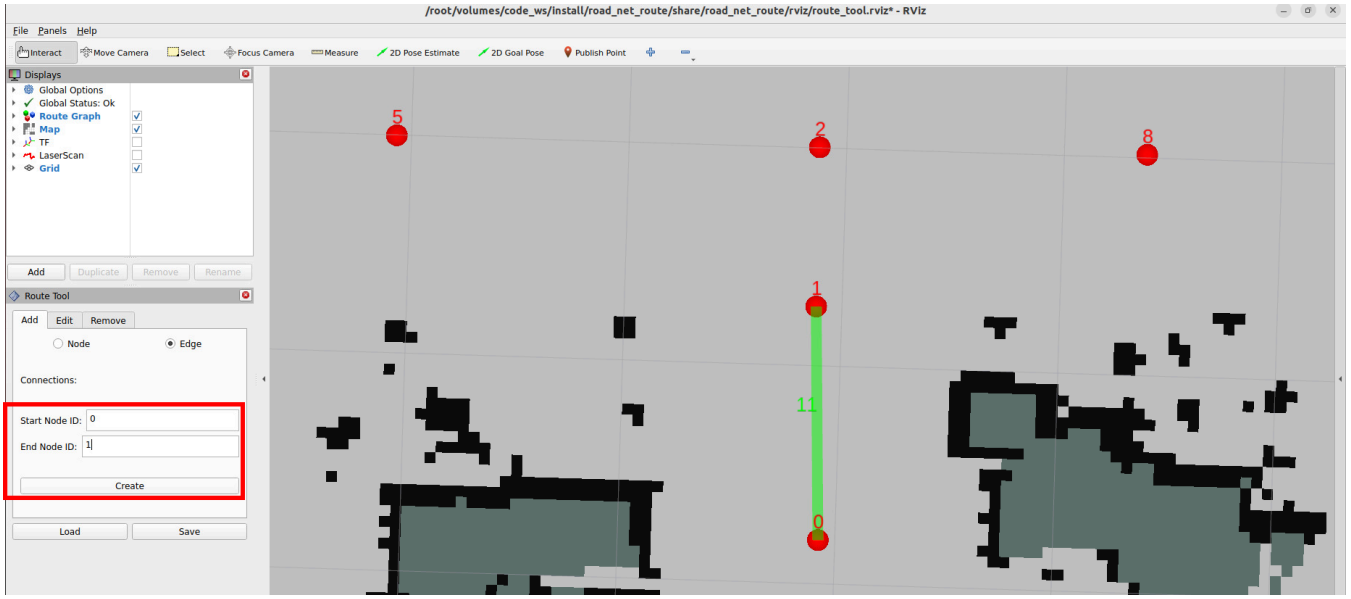
3.2 Adding Edges

- Select Edge in the `Route Tool` tab



- Fill in the starting waypoint of the edge in `Start Node ID`, fill in the ending waypoint of the edge in `End Node ID`, and then click Create to create an edge between the waypoints.

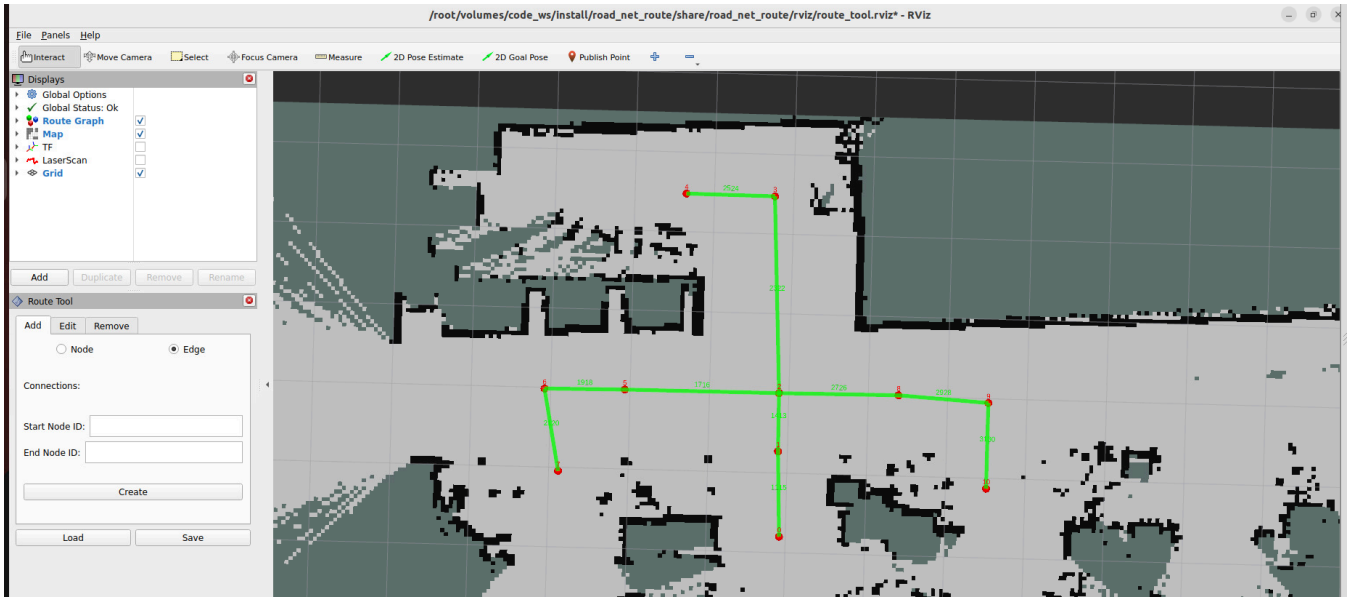
> **Important**
>
> **Note!**
>
> **The road network is a topological structure composed of points and edges, and edges have direction.** For example, the edge from 0→1 created here is one-way. If a two-way edge is needed, an edge from 1→0 must also be created.



- After that, we create the edge from 1-0 so that waypoints 0 and 1 can be traveled in both directions.

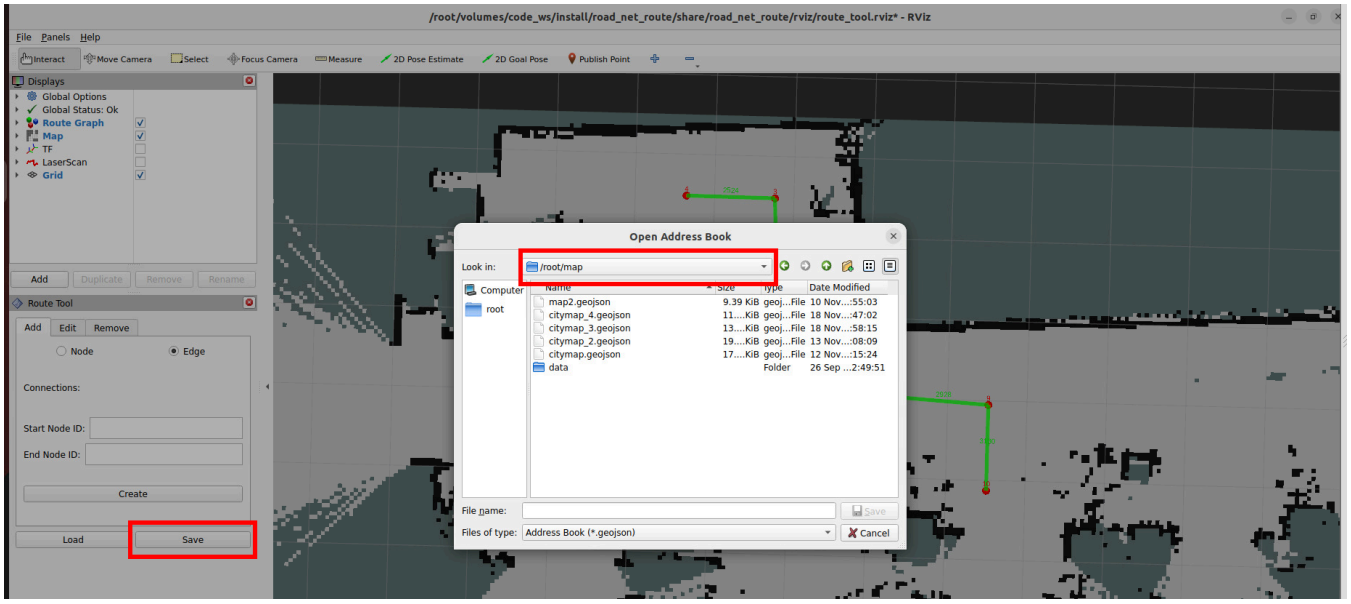


- Repeat the above steps to complete the creation of all edges (whether an edge is set as one-way or two-way is determined by the actual scenario requirements). In this case demonstration, all are set as two-way edges.

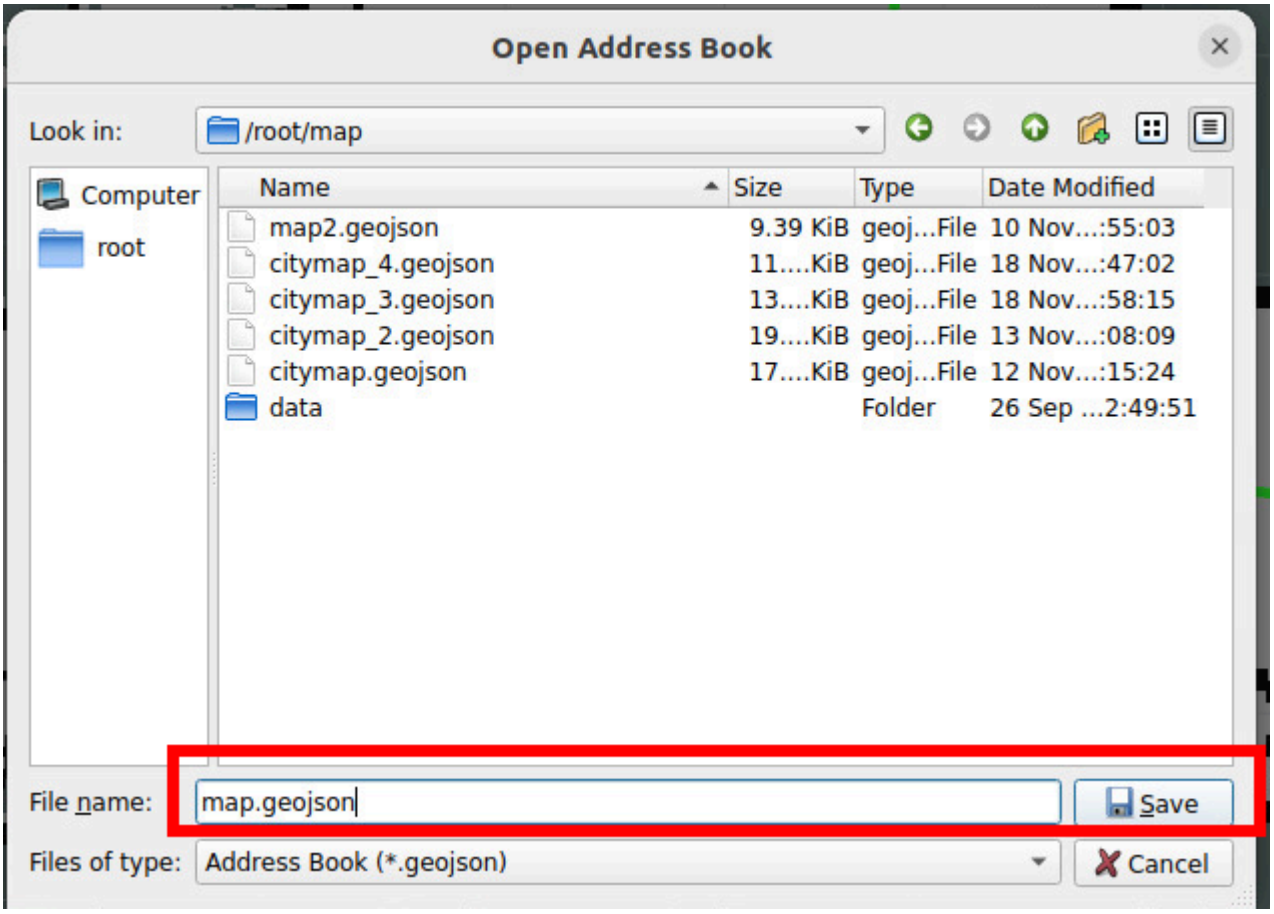


4. Saving the Road Network Description File

- Click save in the `Route Tool` tab, then select the path where the file is to be saved in the pop-up window. Here it is saved in the `$HOME/map/` folder.



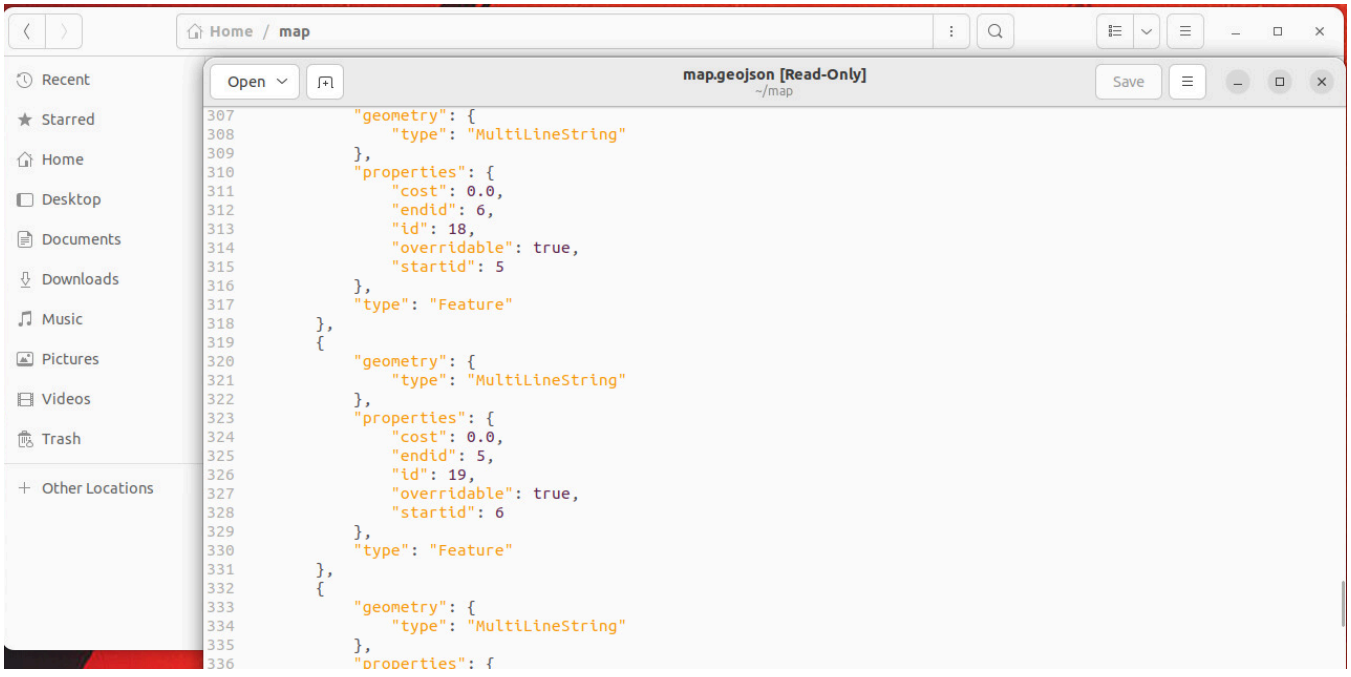
- Then fill in the file name in File name. It is recommended to keep it consistent with the grid map name for easy distinction; the file extension must be `.geojson`. Finally, click save to save the road network description file.



- After a successful save, the terminal will print a prompt message

```
[rviz2-1] [WARN] [1763543213.881559940] [route_tool_node]: The graph filepath was not provided and no default was specified. Failed to save the route graph.
[rviz2-1] [INFO] [1763543726.803456606] [route_tool_node]: Save graph to: /root/map/map.geojson
[rviz2-1] [INFO] [1763543726.803604931] [route_tool_node]: Saving graph file from /root/map/map.geojson, by parser nav2_route::GeoJsonGraphFileSaver
```

- After that, we can directly view the map.geojson road network description file from the host computer's `$HOME/map` folder. The file contains a JSON-format description of the waypoint and edge information.



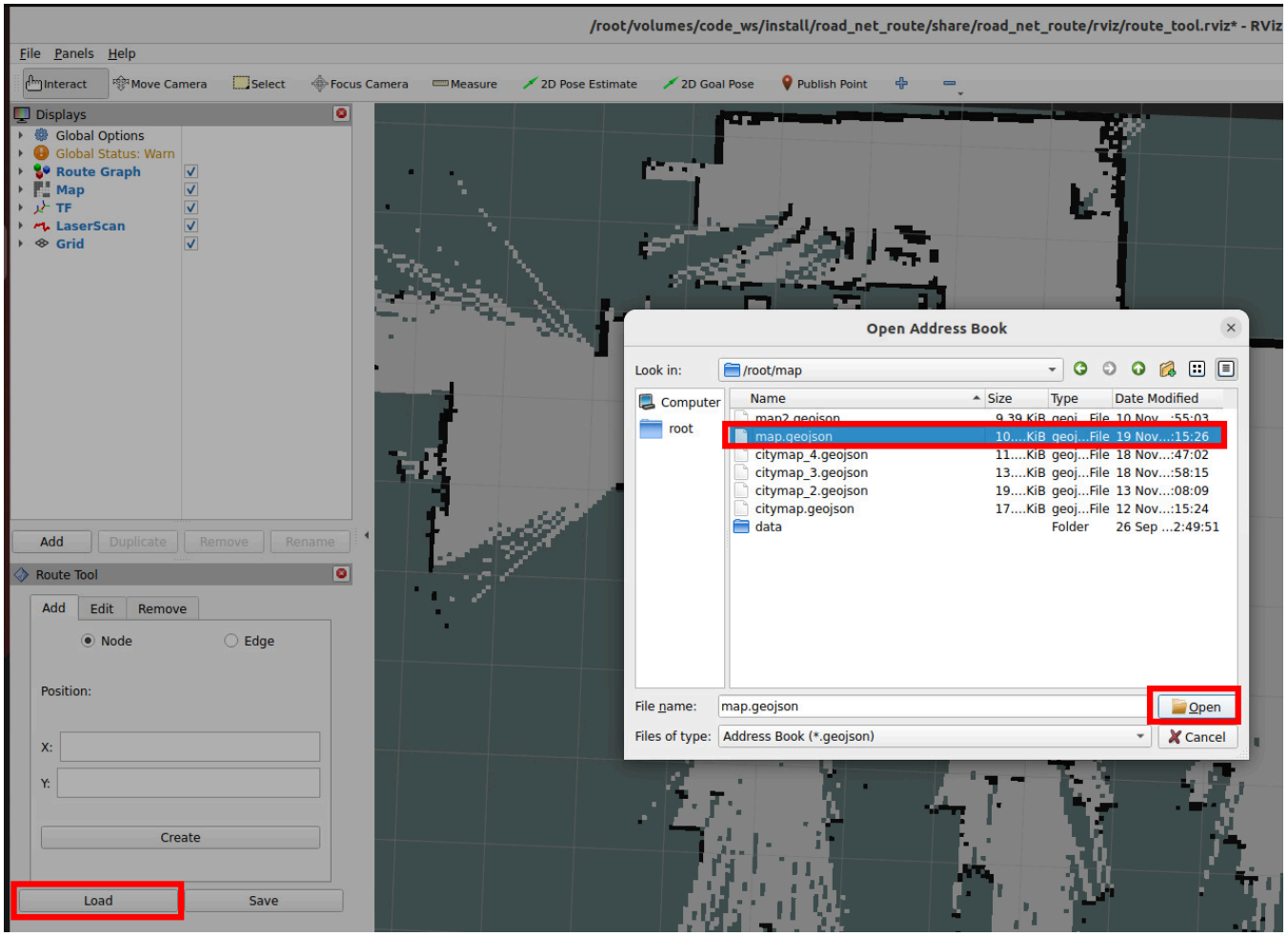
5. Loading and Modifying the Road Network Description File

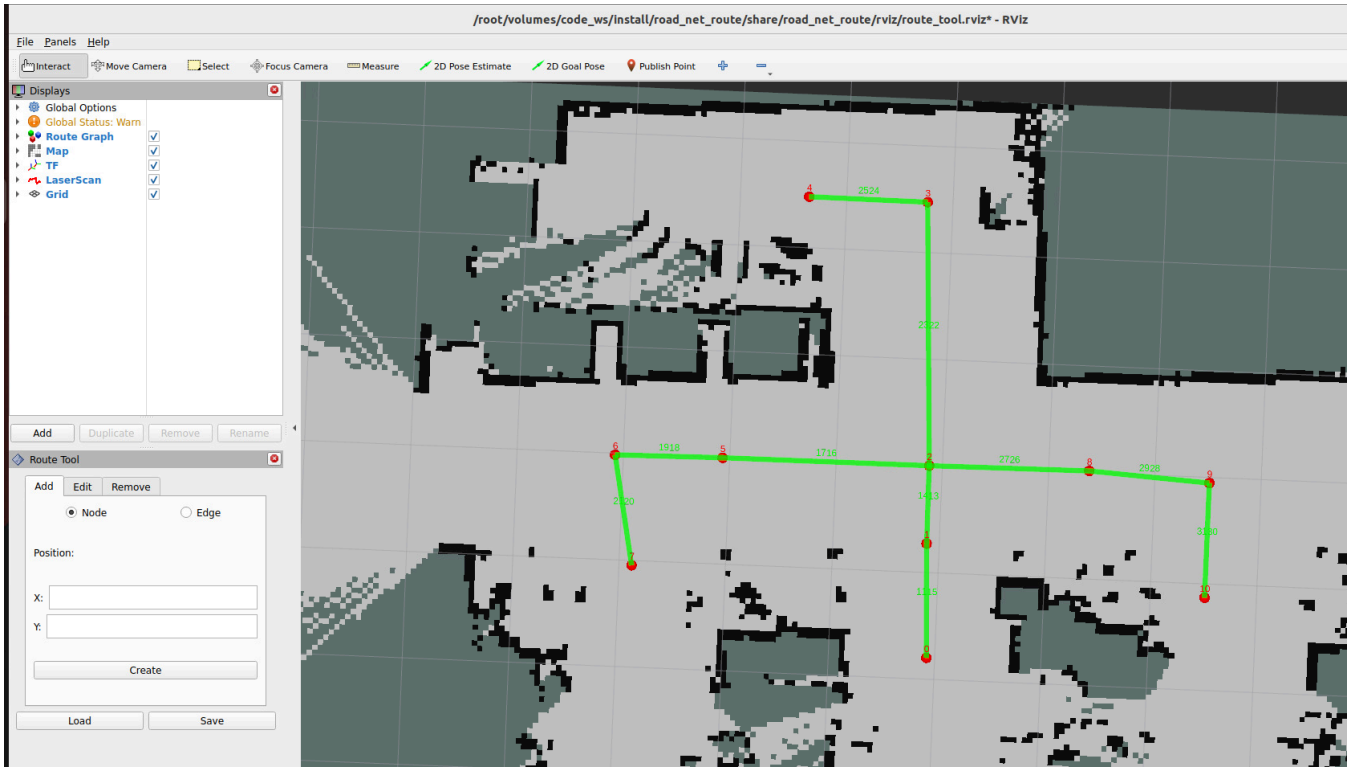
5.1 Loading the Road Network Description File

- If the previous road network description file needs to be modified later, for example to add, remove, or edit points and edges, you can load the previous road network description file for editing.

```
ros2 launch yahboomcar_bringup nav2.launch.py rviz_config:=roadnet_edit
localization_mode:=slam_toolbox
```

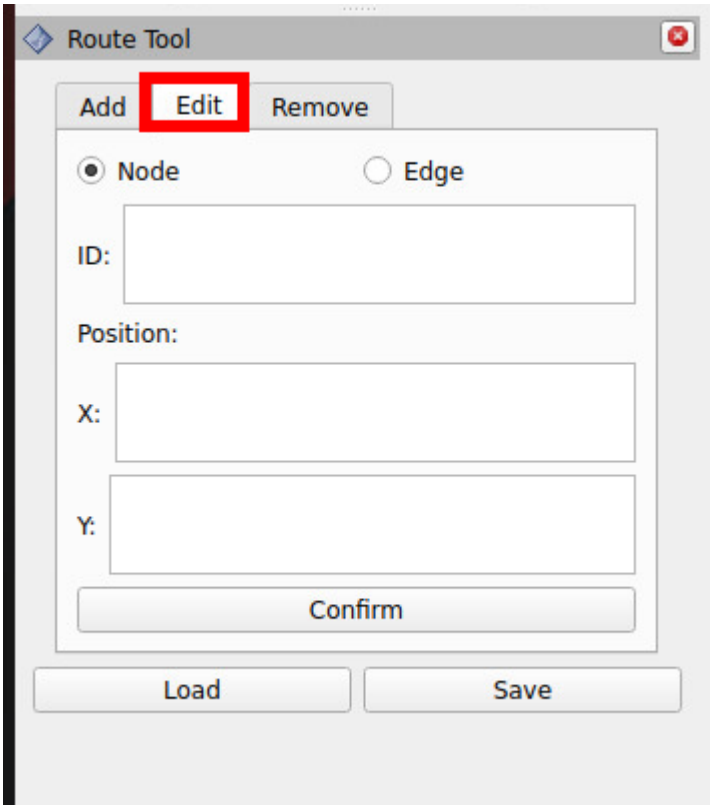
- The **yaml_filename** parameter is the path of the grid map file of the road network to be annotated



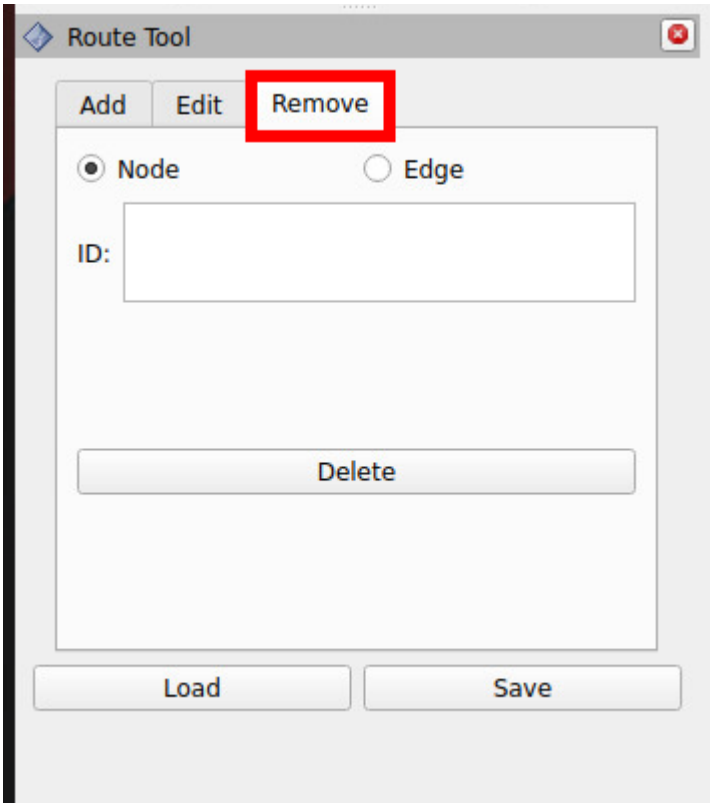


5.2 Editing the Road Network

- If you need to add new waypoints and edges, the operation is the same as the previous steps.
- Click `Edit` to re-edit the information of an existing waypoint or edge.



- Click `Remove` to remove an existing waypoint or edge.


Route Tool
Add
Edit
Remove
Node
Edge
ID:
Delete
Load
Save